

Augmenting Sculpture with Immersive Sonification

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ABSTRACT

We present an artistic Mixed Reality (MR) system that remixes a sculptural element of a building and its aesthetic context to provide an on-site augmented art experience. Mainstream MR systems, particularly art-related, focus on the use of visuals in presenting additional information, whereas the use of audio as the main information channel has rarely been considered. In this work, we explore two different versions of a sonic experience for walking through an artistic staircase to enhance its public's engagement. Our user evaluation reveals the effectiveness of sonic design for a rewarding MR experience. With this, we emphasise the importance of sonic design in MR applications.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Applied computing—Arts and humanities—Sound and music computing

1 INTRODUCTION

Mixed Reality (MR) has been used as a platform to enhance the public's engagement with art [1, 3]. By overlaying additional visual information such as images or texts on top of the real-world features, one can obtain an augmented experience while viewing artworks. However, the use of audio as a dimension to support or complement such user experiences has rarely been considered.

Recently reported in [2, 4], embedding sonic information in MR experiences can be beneficial in various ways. First, it can act as a complement or in substitution for the visual modality in providing information, particularly in certain scenarios where the visual presentation is limited. Second, having multi-dimensions enabled in the MR environment can achieve a natural user interface and thus enhance the overall quality of the experience.

Motivated by such, we create an MR application that provides an augmented experience of a sculptural element of a building and that engages with users in a way that particularly emphasises their sonic dimension of experience. The sculptural element is an elaborate, flying staircase inside of an office building and our application is called the "Sonic Sculptural Staircase". Two different versions of a sonic experience are designed, an "informational design" and an "experiential design", with one using a narrative audio guide and the other using interactive sonic features to prompt users' engagement respectively. Our work is implemented in the Microsoft HoloLens-2 (HL2) [5] and uses an "invisible hologram" approach described in [4] to enable on-site sonic experiences. With the participation of users recruited from the inhabitants of the building, we have formally evaluated these two sonic experiences. Our findings reveal the effectiveness of sonic design for an augmented sculpture experience.

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Figure 1: Left: A user in our MR application with the Microsoft HoloLens 2 and showing some visual holograms. Right: The user is experimenting with a feature ("Coffee Time") in the experiential design condition.

2 SYSTEM OVERVIEW

Our system aims to provide an audio-focused augmented experience of a sculptural feature using HL2. It was specifically built for a staircase located inside a recently constructed building at a university. This staircase was chosen as a suitable sculptural element to be modelled based on a number of artistic and technical considerations.

Artistically, the staircase is a spectacular feature of the multi-storey foyer of an office building that links two departments of a university. Its sculptural form is not only of high aesthetic value, but it has been designed with the underlying rationale of bringing the two departments together. It is frequently used by staff and students of both departments in their daily routine. Physically, the size and spectacular nature of the staircase motivate users to walk around and "through" an MR simulation. In regards to the technical implementation of our system in the HL2, this staircase is located inside of a building where its lighting conditions are relatively stable.

2.1 Design

Two versions of a sonic experience are designed in our system: "informational" and "experiential" designs. Each design reflects on different uses of audio to provide additional information of the staircase's aesthetic background. In addition to their sonic layers, there were minimal sets of visual overlays designed in these experiences to respond to user interactions, mapping to the theme of this staircase and the building in which it is situated. The user is asked to engage with these sonic experiences by either walking or hand interactions in HL2 (See Fig. 2). Both versions were designed to have a similar 5 minutes time length with a consideration of balancing conditions in user evaluation.

2.1.1 Informational Design

This sonic experience is delivered by a narrative audio guide with virtual visual hints to provide background information about the selected sculptural staircase. The audio guidance includes four major parts, allowing the user to walk around and trigger related audio content to access background information. During this experience, the user is encouraged to look at different parts of the staircase and its surroundings to understand its aesthetic context.

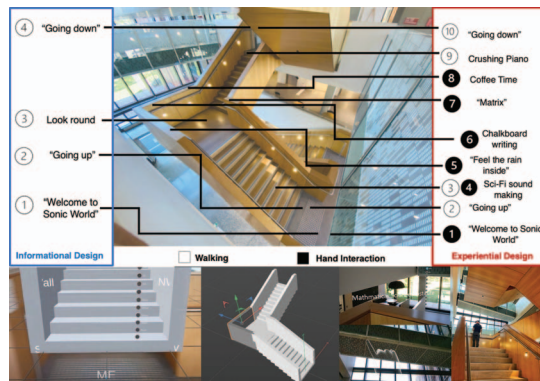


Figure 2: Up: Features map for both design versions. Down: the staircase model (left) maps to real-world (middle); A user enjoying a feature in the informational design condition (right).

2.1.2 Experiential Design

This sonic experience requires the user to actively engage with the staircase to obtain its aesthetic context. Ten different interactive sonic features are embedded, mainly comprising indirect audio information such as field recordings of the building or music pieces with specific themes related to the staircase's aesthetic context. During this experience, the user is asked to explore features either by walking up and around the staircase or hand gesture interactions at the middle-level platform to obtain the related information of the staircase. By exploring these features, the user is able to stand at different parts of the staircase and have different viewing angles to appreciate its design.

2.2 Technical Implementation

Both designs were realised on a HL2 and programmed in Unity version 2019.3.7.f1. Microsoft's Mixed Reality Toolkit (MRTK) [6] and World Locking Tool (WLT) [8] were used to achieve real-world location-specific interactive features about the 3D staircase.

To enable location-specific features, an (invisible) 3D staircase model was created (See Fig. 2) and used as the reference to model the relationship between the user's HMD and the real-world staircase in Unity. The 1:1 scale staircase model was manually mapped to the real-world staircase using virtual spatial markers through a UI panel while running the WLT application in HL2. The WLT engine locks the staircase model in the specific physical environment and can persist every time the application is restarted [7].

All sonic features were built based on the staircase hologram with each feature represented as a child hologram. These holograms were applied as Game Objects in Unity for the development. In particular, features containing visual overlays and interaction were implemented with MRTK and a collision detection algorithm written in C#, while the sound used in the experience, consisting of field recordings and synthesised sounds from the Disunity library [9], were made with the AudioSource component in Unity.

3 USER EVALUATION SPOTLIGHT

To test our system, we have evaluated two versions of sonic experiences with 10 participants (3 female, 7 male) who worked or studied in the building containing the staircase. None of the participants had extensive prior experience in the use of a Microsoft HoloLens. All participants said that they regularly used the part of the staircase in their daily lives; on average they reported that they used this part of the staircase 16 times during a week.

Our evaluation consisted of an independent measures study with one condition (MR system design) that had two levels (informational design and experiential design). The user study design had

two components: first, a 7-Likert scale questionnaire with six criterion examining the basic functionality of the built prototypes; and secondly, semi-structured interviews including three questions to gather further context about their experience. The study was conducted immediately after participants finished using our MR system.

Functionally Engaging All six criterion examined in the questionnaire: 1) *Ease of Use*; 2) *Enhanced Understanding*; 3) *Involvement*; 4) *Pleasure*; 5) *Engagement Through Exploration*; 6-a) *Flexibility of Exploration*; 6-b) *Relates to Experience* received positive ratings greater than neutral across both designs. This shows our system to be functional and allow the user to engage with the staircase in a meaningful way. Of these criterion, participants gave highest rating to *Pleasure*. Moreover, the later interview study results elaborated reasons for their excitement on both design versions: "a totally different staircase experience/never ever have before", "being able to explore the space", "the experience is straightforward, easy to navigate" and "sound complements the feeling of presence".

The Balance of Sonic Design to Provide Information Participants provided suggestions to the sonic design in our system on both design versions in the interview study. For informational design, comments were clustered to the audio guide: "more clear information about how long the experience lasts and listening options", and "make the audio more spatial". For experiential design, comments were clustered to the field recordings and electronic music: e.g., "adjustment of piano scale" and "make the sound more prominent to the building environment". A participant also commented on desiring more straightforward guidance in this design.

More Visuals as Substitutions Participants from both design versions desire more (concrete) visuals to complement their sonic experiences: "using visual guidance to complement audio information/sound" and "visual is a bit abstract".

4 CONCLUSION

We created an artistic MR system that contains two different sonic experiences for a sculptural staircase. Our user evaluation highlighted the effectiveness of sonic design in MR applications. We suggest the findings as a groundwork to similar multimodal applications.

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